

Skills

Languages: C++ (11/14/17/20, Template Metaprogramming), Erlang, SQL, MATLAB, Python

Technical Foundations: Data Structures, Algorithms, System Design, HLD, LLD, Design Patterns, Linux, IPC, Makefiles, GDB, Multithreading, Lock-free Data Structures, Synchronisation Primitives, OpenSSL, Socket Programming, Event-Driven Architectures, Distributed Systems, Microservices, REST APIs, Kafka, PostgreSQL, NoSQL

Ecosystem & Infrastructure: Docker, Kubernetes, Containerization, CI/CD pipelines, GitHub Actions, ElasticSearch, Prometheus, Grafana, Log Analysis, perf, Postman, Vim, Git, Agile, JIRA

Experience

Staff Backend Engineer, GreyOrange, Gurgaon

Jan 2023 – Nov 2024

- Engineered fault-tolerant, high-throughput features for large-scale distributed **C++** and **Erlang** backend systems, ensuring **99.99% uptime** and site reliability under heavy concurrency and production load.
- Solved recurring data corruption incidents by designing and developing a data sanity module and real-time APIs for monitoring data integrity, **reducing incidents by 45%** and cutting debugging overhead by 30 engineer hours/week.
- Implemented a lock-free SPMC queue using synchronization primitives in **C++** for task scheduler service, reducing contention and increased scheduling efficiency by **70%**
- Diagnosed and resolved **100+ production incidents** by analyzing server logs, yielding quick RCA, resolving bugs, implementing hotfixes, backports and 50+ unit/integration tests, within defined SLAs, to eliminate top causes of crashes.
- Re-architected performance-critical APIs by improving cache policies and simplifying request routing logic, **accelerating response times by 30%**; utilized **Kafka** to develop and containerize event-driven features, and collaborated with Product Manager and downstream teams to design new topics and payload structures.
- Mentored** junior engineers, delivered technical training sessions, raising backend competency across the team.

Senior Software Engineer, PineLabs, Noida

Aug 2021 – Dec 2022

- Designed and developed features and enhancements to the PLUTUS-India and All-Tap application, improving **transaction processing speed by 15%** for over 10,000 users.
- Automated merchant settlements with a multithreaded C++ service using thread pools, cutting daily ops time to **zero**.
- Modernized legacy payments by migrating to Android POS systems, **improving efficiency by 8%**; **Optimized database performance** by writing migrations to add indexes, and materialized views.
- Built a C++ log parser service with efficient string handling to auto-generate diagnostic reports for top-occurring errors, significantly reducing incident frequency; introduced Test Driven Development with GSuite to achieve >95% coverage and **cut production defects by 35%**.
- Improved system reliability and ensured 99.99% transaction success by resolving **50+** production bugs and socket communication issues using **core dump analysis** and **ThreadSanitizer** to uncover hidden races and memory errors.
- Led **cross-functional collaboration** with infrastructure and IT teams to set up a dedicated development environment, while gaining deep domain expertise in banking payments through hands-on work with distributed backends, **REST APIs**, and **socket programming**.

Associate Software Developer, MathWorks, Bangalore

Oct 2018 – Apr 2021

- Accelerated developer workflows by building MATLAB CLI automation tools used by **250+ engineers**, streamlining production environment setup.
- Improved Simulink search accuracy to **95%** by designing search algorithms and efficient string-processing modules.
- Reduced technical debt and improved diagnostics by deprecating legacy APIs, migrating to modern replacements, and applying the **Adapter Design Pattern** to integrate a third-party library into Simulink's error reporting system, increasing code maintainability and boosting test coverage by **25%**.

Software Developer, Dassault Systèmes, Pune

Jan 2017 – Sep 2018

- Delivered new CAD functionalities in the Product Structure Design team with strong emphasis on **computational geometry** and Microsoft COM development.
- Implemented optimized n-ary tree algorithms with the Factory Design Pattern for product structure selection & copy-paste, improving performance and lowering memory use.
- Improved 3DEXPERIENCE CAD stability and runtime performance by applying C++ design patterns to simplify complex object hierarchies, and resolving **15+ critical memory leaks and bugs**.

Education

National Institute of Technology Surat, India

Graduated: May 2016

B.Tech. in Electronics and Communication Engineering